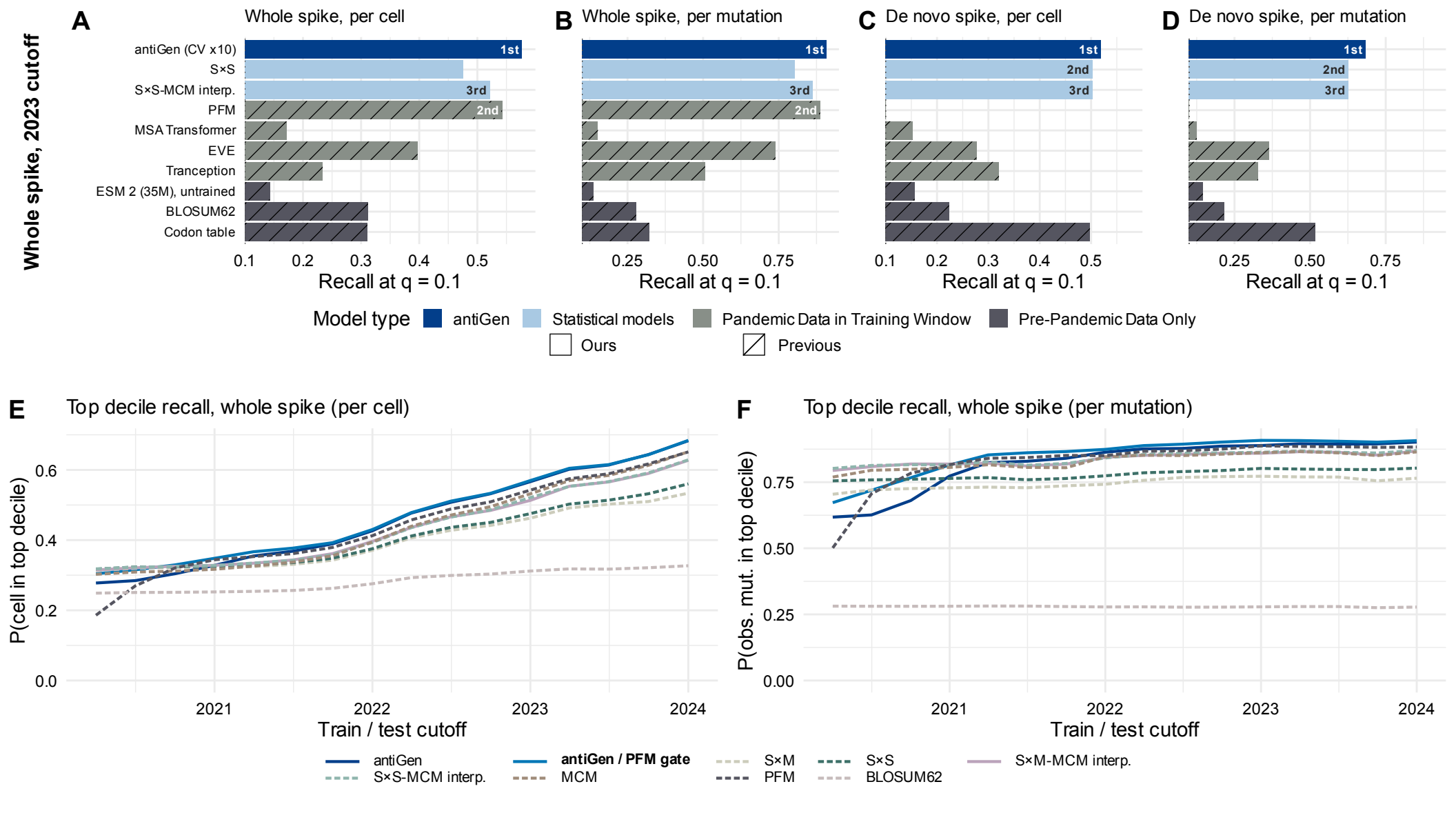
A
Whole spike, per cell
B
Whole spike, per mutation
C
De novo spike, per cell
D
De novo spike, per mutation
Whole spike, 2023 cutoff
antiGen (CV x10)
S×S
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
ESM 2 (35M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1
0.2
0.3
0.4
0.5
0.25
0.50
0.75
Recall at q = 0.1
Model type
antiGen
Statistical models
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
E
Top decile recall, whole spike (per cell)
P(cell in top decile)
0.0
0.2
0.4
0.6
2021
2022
2023
2024
Train / test cutoff
F
Top decile recall, whole spike (per mutation)
P(obs. mut. in top decile)
0.00
0.25
0.50
0.75
antiGen
S×S-MCM interp.
antiGen / PFM gate
MCM
S×M
PFM
S×S
BLOSUM62
S×M-MCM interp.